Exercises (Algebra-1)

October 9, 2026

1. Let G be a finite group, with exactly one non-identity element f such that $f^2 = e$. Prove that
$$\prod_{g \in G} g = f.$$

2. Let n be a positive integer. Prove that $F\{x, y\}$ contains a subgroup which is isomorphic to $F(S)$, where $F(S)$ is a free group on a set S having n elements.

3. Let m and n be positive integers. Prove that there is a surjective morphism of groups from F_n onto F_m if and only if m is less than or equal to n. (**Note**: Here F_n denote the free group on a set which have n elements.)

4. Check in detail that $\{x, y \mid xyx^{-1}y^{-1}\}$ is a presentation for $\mathbb{Z} \times \mathbb{Z}$.

5. Which of the following define morphisms of groups from $(\mathbb{C} \setminus \{0\}, \cdot)$ to $(\mathbb{C} \setminus \{0\}, \cdot)$?

   (a) $z \mapsto z^*$

   (b) $z \mapsto z^2$

   (c) $z \mapsto iz$

   (d) $z \mapsto |z|$

6. Show that a function $\varphi : G \to G'$ is a morphism of groups if and only if
$$\{(g, \varphi(g)) \mid g \in G\}$$
is a subgroup of $G \times G'$.

7. Prove that $(\mathbb{C}, +)$ is isomorphic to $\mathbb{R} \times \mathbb{R}$, and that $(\mathbb{C} \setminus \{0\}, \cdot)$ is isomorphic to $\mathbb{R}^+ \times \mathbb{C}$.

8. Call H a proper subgroup of the group G if H is neither $\{e\}$ nor all of G. Find a group which is isomorphic to one of its proper subgroups.

9. Show that $(\mathbb{R}, +)$ is not isomorphic to $(\mathbb{Q}, +)$ and that $(\mathbb{R} \setminus \{0\}, \cdot)$ is not isomorphic to $(\mathbb{Q} \setminus \{0\}, \cdot)$. Is $(\mathbb{R}, +)$ isomorphic to $(\mathbb{R} \setminus \{0\}, \cdot)$?

10. Let $D_n = \{r, s \mid r^n, s^2, srs^{-1}r\}$ denote the dihedral group of order $2n$, where $n \geq 2$. Find the subgroup of D_n generated by r^2 and r^2s, distinguishing carefully between the cases n odd and n even.

11. The group defined by generators a, b and relations

$$a^8 = b^2a^4 = ab^{-1}ab = e$$

has order at most 16.

12. Show that the group defined by generators a, b and relations $a^2 = e,\ b^3 = e$ is infinite and non-commutative.

13. Let $(M, p, 1)$ be a monoid and let $m \in M$. Define a new product p_m in M by

$$p_m(a, b) = amb.$$

Show that this defines a semigroup. Under what condition on m do we have a unit relative to p_m?
(**Note:** Here p is the law of composition of M.)

14. Give an example of a function $f : X \to Y$ and a subset A of X for which A is properly contained in $f^{-1}(f(A))$.

15. Show that any group G in which every $a \in G$ satisfies $a^2 = e$ is commutative. What if $a^3 = e$ for every a?

16. How many different equivalence relations can be defined on a set of five elements?

17. Show that the three groups described below are isomorphic.

    (a) $G = \langle a, b \mid a^2 = e,\ ab = b^{-1}a\rangle$.

    (b) $H = \langle s, t \mid s^2 = t^2 = e\rangle$.

    (c) $D = \{$distance-preserving permutations of the set $\mathbb{Z}\}$, i.e., the group consisting of all translation-maps $n \mapsto n+c$ $(c \in \mathbb{Z})$ and all reflection-maps $n \mapsto -n + d$ $(d \in \mathbb{Z})$.

18. Let $f : X \to Y$ be a function and $F : \mathcal{P}(Y) \to \mathcal{P}(X)$ be defined by

$$F(A) = f^*(A) \quad \text{for all } A \subseteq Y.$$

Then, prove the following.

    (i) f is injection if and only if F is surjection.

    (ii) f is surjection if and only if F is injection.

    (iii) f is a bijection if and only if F is a bijection.

19. Let X be a set and define $f : \mathcal{P}(\mathcal{P}(X)) \to \mathcal{P}(X)$ by

$$f(C) = \bigcup_{A \in C} A \quad \text{for any } C \subseteq \mathcal{P}(X).$$

Then, find two distinct right inverses of f.

20. A pointed set is a pair (S, x) with S a set and $x \in S$. A morphism of pointed sets $(S, x) \to (S', x')$ is a triple (f, x, x'), where $f : S \to S'$ is a function such that $f(x) = x'$. Show that pointed sets form a category.

21. Let $\varnothing \neq A \subseteq X$ and $f : A \to Y$ be a function. Does there exist a function $g : X \to Y$ such that $g|_A = f$? If yes, how many such functions g can be found?

22. If $f : \mathbb{R} \to \mathbb{R}$ is defined by $f(a) = a^2$ for all $a \in \mathbb{R}$, then determine

$$f^*([-2, 8)), \quad f^*((-\infty, 0]), \quad f^*((-1, 1)) \quad \text{and} \quad f^*(\mathbb{Z}).$$

23. Let N denote the set $\{1, 2, 3, \ldots\}$ of natural numbers, and let $s : N \to N$ be the shift map, defined by $s(n) = n + 1$. Prove that s has no right inverse, but that it has infinitely many left inverses.

24. Let x, y, z, and w be elements of a group G.

    (a) Solve for y, given that $xyz^{-1}w = e$.

    (b) Suppose that $xyz = 1$. Does it follow that $yzx = e$? Does it follow that $yxz = 1$?

25. Let G be a group. Define an opposite group G° with law of composition $a * b$ as follows: The underlying set is the same as G, but the law of composition is $a * b = ba$. Prove that G° is a group.

26. Prove that a group G and its opposite group G° (Exercise 25) are isomorphic.